

**Group:** \_\_\_\_\_ **Section:** \_\_\_\_\_

**Member 1:** \_\_\_\_\_ **BUID:** \_\_\_\_\_

**Member 2:** \_\_\_\_\_ **BUID:** \_\_\_\_\_

**Member 3:** \_\_\_\_\_ **BUID:** \_\_\_\_\_

**Member 4:** \_\_\_\_\_ **BUID:** \_\_\_\_\_

## Project 2 Real Data Analysis

### Purpose

Today you will apply the Project 2 inference comparison to a real data set and connect real-data diagnostics to the simulation study.

### Shared Group Workflow

Work in one shared Google Doc and one shared analysis folder. Assign the four roles below, then have every member review the real-data analysis and its connection to the simulations.

- **Member 1 – Data lead:** import, inspect, and document the data and variables.
- **Member 2 – OLS lead:** fit the planned model and interpret the target coefficient or prediction.
- **Member 3 – Bootstrap lead:** compute bootstrap intervals and compare them with classical intervals.
- **Member 4 – Diagnostics editor:** connect assumptions, diagnostics, and conclusions in the shared report.

### In-Class Work

- (A) **Load and explore.** Import the real data set and inspect the response, predictors, missing values, and variable types.
- (B) **Fit OLS.** Fit the planned regression model and interpret the main coefficient or prediction target.
- (C) **Classical versus bootstrap.** Compare classical and bootstrap intervals for the target quantity.
- (D) **Diagnostics.** Check residual patterns and connect them to assumptions from the simulation design.
- (E) **Check-in.** Do the real-data diagnostics resemble one of your simulation scenarios?

### Deliverable

Submit the Project 2 progress report through Gradescope or Blackboard by Sunday, November 1 at 10:00 PM. Include the real-data analysis, diagnostics, shared Google Doc link, and reproducible code files.